



Installing programs on Linux can be really difficult, especially if you are accustomed to the Windows way. Unlike Windows, Linux doesn't have wizards that automate the task—you have to use some command-line utility. At most, you will be provided with a graphical front-end to the command. Moreover, each Linux distribution follows a different package format, such as tar, rpm, deb, etc. In spite of all these apparent setbacks, you will be surprised to find people swearing by Linux package installers. The reason for this being that they do a very clean job—no registry settings to be tweaked.

In this workshop, we will cover the basic rpm command-line tool, a graphical

utility Gnome-RPM, the default package managers of Red Hat and SuSE, and finally, a common package manager for both distributions—Nautilus.

Red Hat Package Manager

Red Hat Package Manager is a powerful command-line package management system tool, which is developed by Red Hat. With the help of this tool you can query, install, verify, upgrade, as well as uninstall packages. The files have an .rpm extension, and the filenames follow a standard format of 'package-version-build.architecture.rpm'. An increasing number of distributors are using the rpm format for their programs, as they are easier to install

than the other two methods—source code or pre-compiled binaries.

Querying

The first step, called Querying, involves checking whether the package is already installed or not. To check this, you need to execute the rpm command with the -q option; `'rpm -q <package name>'`. This will show you the details of the package, which will be in the format 'package-version-build'—for example if you issue the command `'rpm -q rpm'` on a machine with Red Hat version 9, the result is 'rpm-4.1-1.06'.

Now, if the package does not exist, you can install it. But, if it is already installed, you can choose to upgrade, providing the package you have is a later version.

Installing

To install a new package you need to execute the rpm command with -ivh option, say, `'rpm -ivh (file name)'`. Here the 'i' stands for install, 'v' for verbose and 'h' for hash progress bar.

When you execute the command you could receive an error: 'Failed dependencies', followed by the packages required. Now, you have to issue a new command, and specify each and every dependency file in the command line. Make sure you have the package files in your current directory—copy the files from the CD-ROM or download from the Internet, accordingly.

Upgrading

If you find a previous version of the package already installed on your system you should upgrade, unless you want two versions of the same package on your system. You do so by using the -U option; `'rpm -Uvh rpm'`

Freshening

To make sure that you are upgrading to a later version of a package, you can use the -F option. Unlike -U, -F won't install the package on the system if the installation version already exists; for instance, `'rpm -Fvh (file name)'`.

Verifying

Once you have installed the package, you can view the information stored by rpm for each file. The information stored consists of file size, ownership, permissions, and MD5 checksum. The command used is `'rpm -V <package name>'`. The result is a